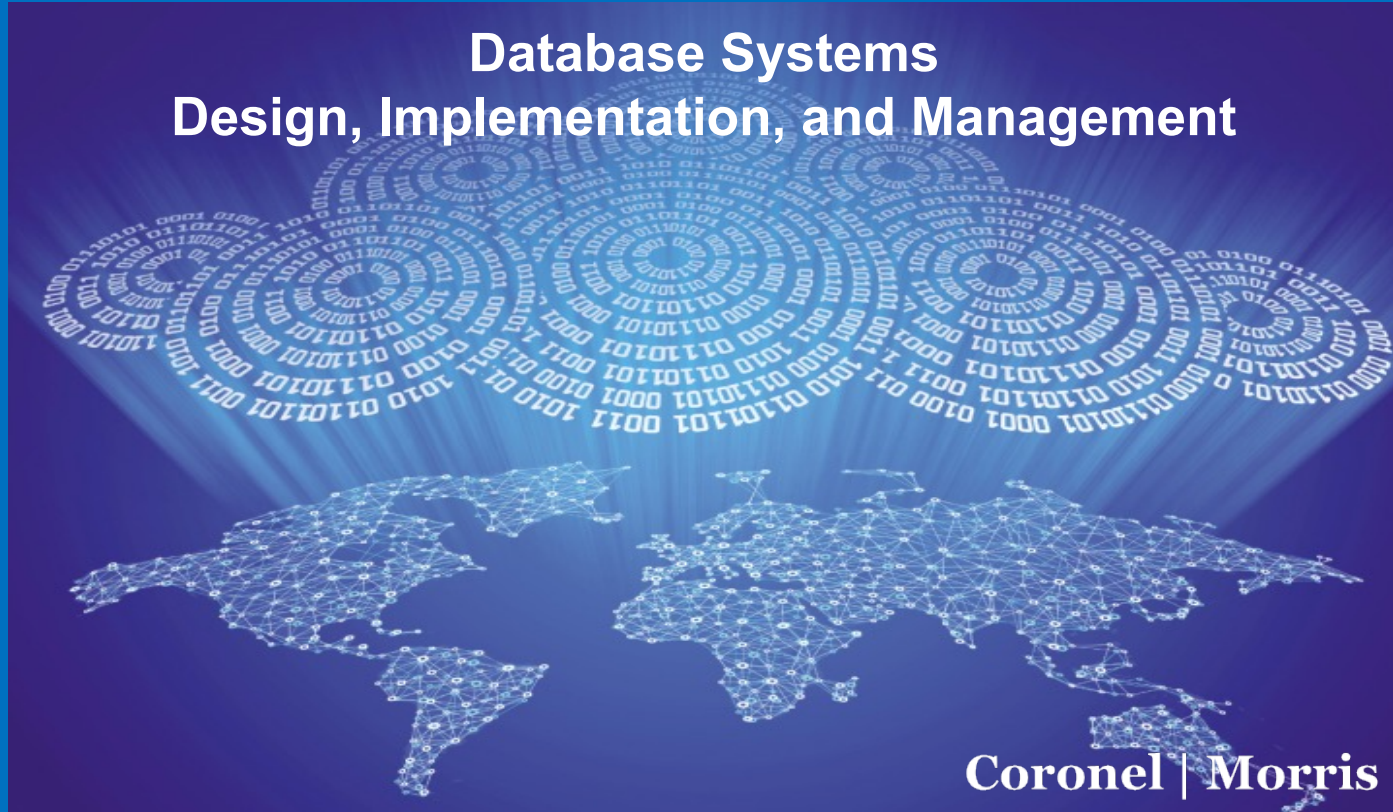


Database Systems Design, Implementation, and Management



Chapter 4 Entity Relationship (ER) Modeling

Learning Objectives

- In this chapter, students will learn:
 - The main characteristics of entity relationship components
 - How relationships between entities are defined, refined, and incorporated into the database design process
 - How ERD components affect database design and implementation
 - That real-world database design often requires the reconciliation of conflicting goals

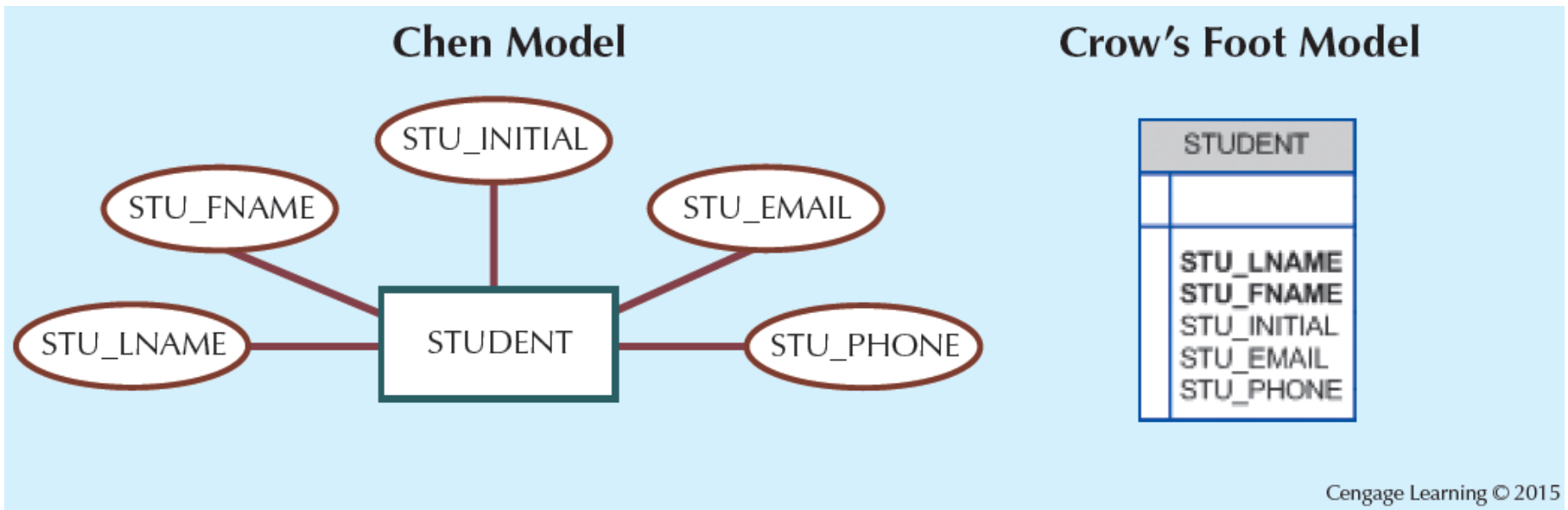
Entity Relationship Model (ERM)

- Basis of an entity relationship diagram (ERD)
- ERD depicts the:
 - Conceptual database as viewed by end user
 - Database's main components
 - Entities
 - Attributes
 - Relationships
- Entity - Refers to the entity set and not to a single entity occurrence

Attributes

- Characteristics of entities
- **Required attribute:** Must have a value, cannot be left empty
- **Optional attribute:** Does not require a value, can be left empty
- **Domain** - Set of possible values for a given attribute
- **Identifiers:** One or more attributes that uniquely identify each entity instance

Figure 4.1 - The Attributes of the Student Entity: Chen and Crow's Foot

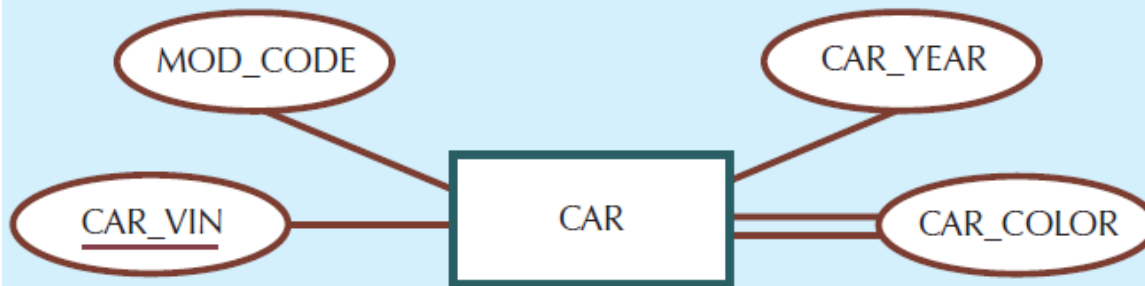


Attributes

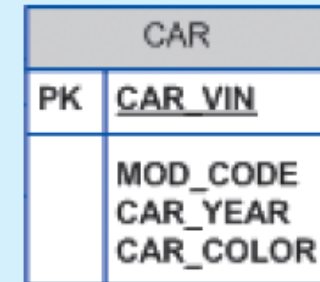
- **Composite identifier:** Primary key composed of more than one attribute
- **Composite attribute:** Attribute that can be subdivided to yield additional attributes
- **Simple attribute:** Attribute that cannot be subdivided
- **Single-valued attribute:** Attribute that has only a single value
- **Multivalued attributes:** Attributes that have many values

Figure 4.3 - A Multivalued Attribute in an Entity

Chen Model



Crow's Foot Model



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Attributes

- **Multivalued attributes:** Attributes that have many values and require creating:
 - Several new attributes, one for each component of the original multivalued attribute
 - A new entity composed of the original multivalued attribute's components
- **Derived attribute:** Attribute whose value is calculated from other attributes
 - Derived using an algorithm

Figure 4.6 - Depiction of a Derived Attribute

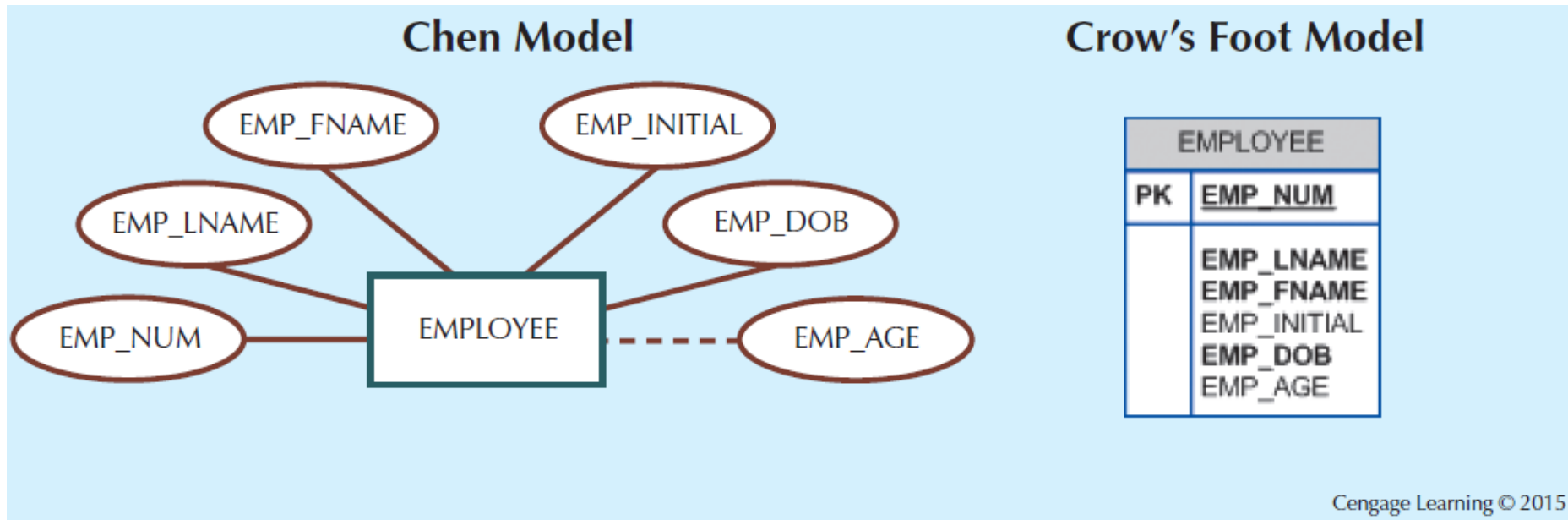


Table 4.2 - Advantages and Disadvantages of Storing Derived Attributes

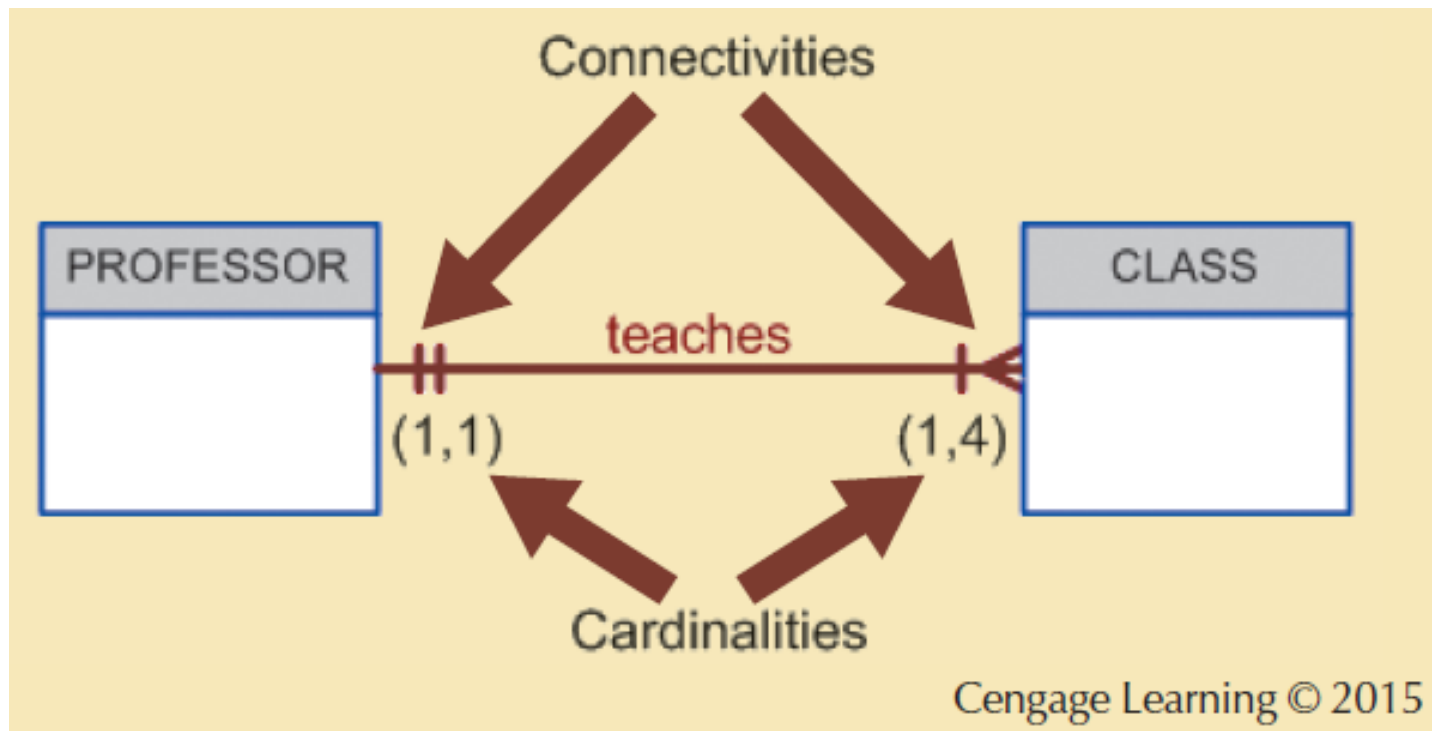
	STORED	NOT STORED
Advantage	- Saves CPU processing cycles - Saves data access time - Data value is readily available - Can be used to keep track of historical data	- Saves storage space - Computation always yields current value
Disadvantage	Requires constant maintenance to ensure derived value is current, especially if any values used in the calculation change	- Uses CPU processing cycles - Increases data access time - Adds coding complexity to queries

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Relationships

- Association between entities that always operate in both directions
- **Participants:** Entities that participate in a relationship
- **Connectivity:** Describes the relationship classification
- **Cardinality:** Expresses the minimum and maximum number of entity occurrences associated with one occurrence of related entity

Figure 4.7 - Connectivity and Cardinality in an ERD



Relationship Strength

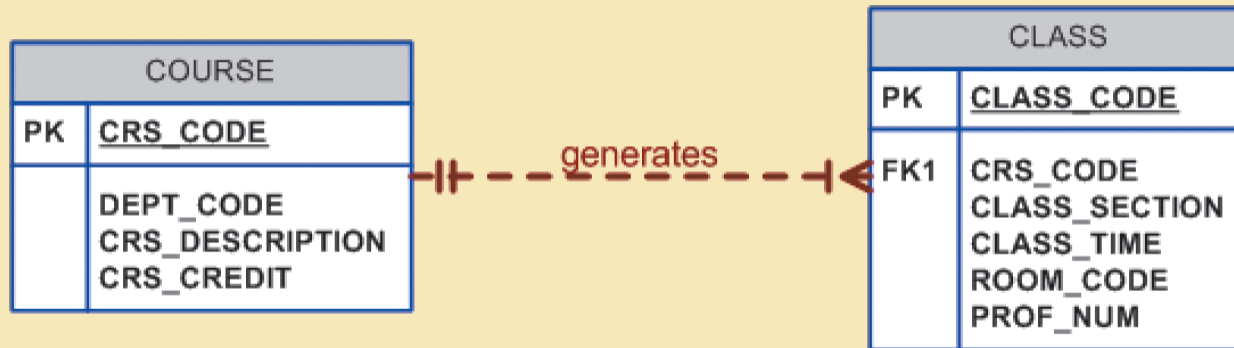
Weak (non-identifying) relationship

- Primary key of the related entity does not contain a primary key component of the parent entity

Strong (identifying) relationships

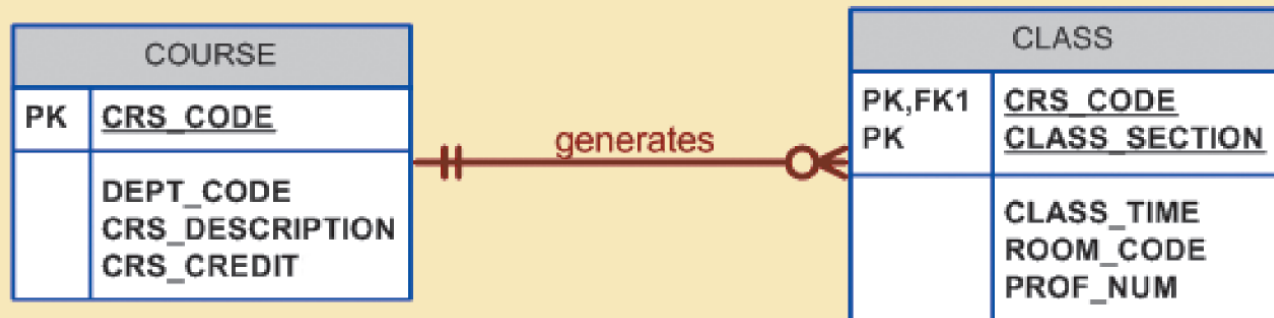
- Primary key of the related entity contains a primary key component of the parent entity

Figure 4.8 - A Weak (Non-Identifying) Relationship between COURSE and CLASS



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Figure 4.9 - A Strong (Identifying) Relationship between COURSE and CLASS



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Relationship Participation

Optional participation

- One entity occurrence does not require a corresponding entity occurrence in a particular relationship

Mandatory participation

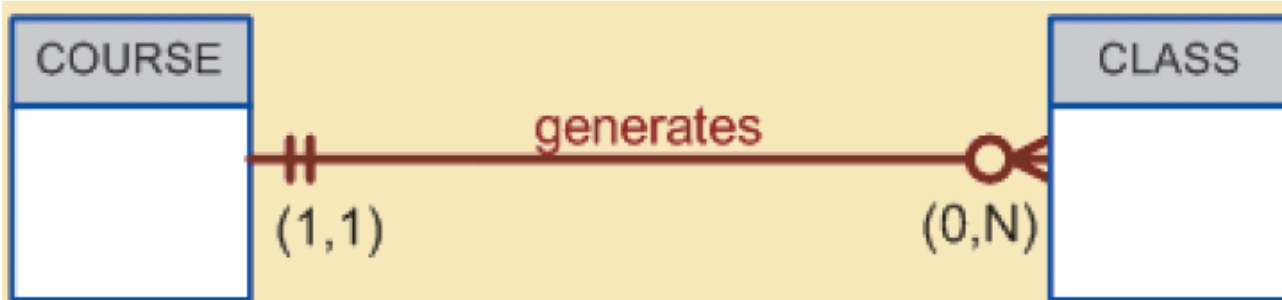
- One entity occurrence requires a corresponding entity occurrence in a particular relationship

Table 4.3 - Crow's Foot Symbols

CROW'S FOOT SYMBOLS	CARDINALITY	COMMENT
	(0,N)	Zero or many; the "many" side is optional.
	(1,N)	One or many; the "many" side is mandatory.
	(1,1)	One and only one; the "1" side is mandatory.
	(0,1)	Zero or one; the "1" side is optional.

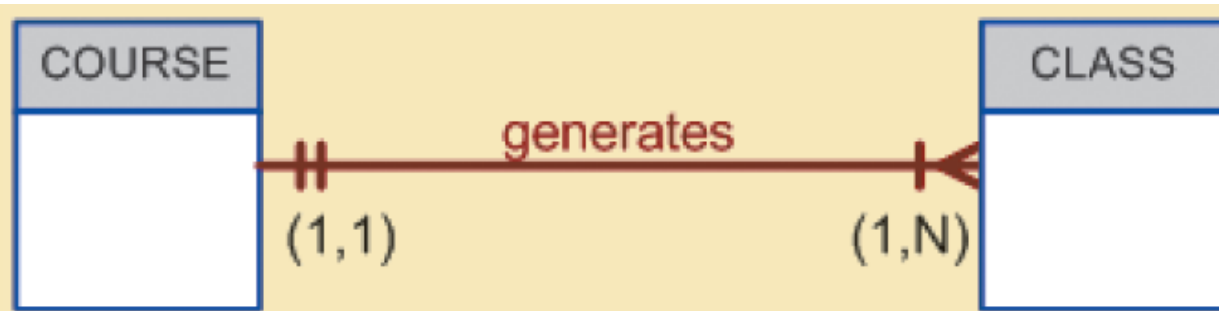
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Figure 4.13 - CLASS is Optional to COURSE



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Figure 4.14 - COURSE and CLASS in a Mandatory Relationship

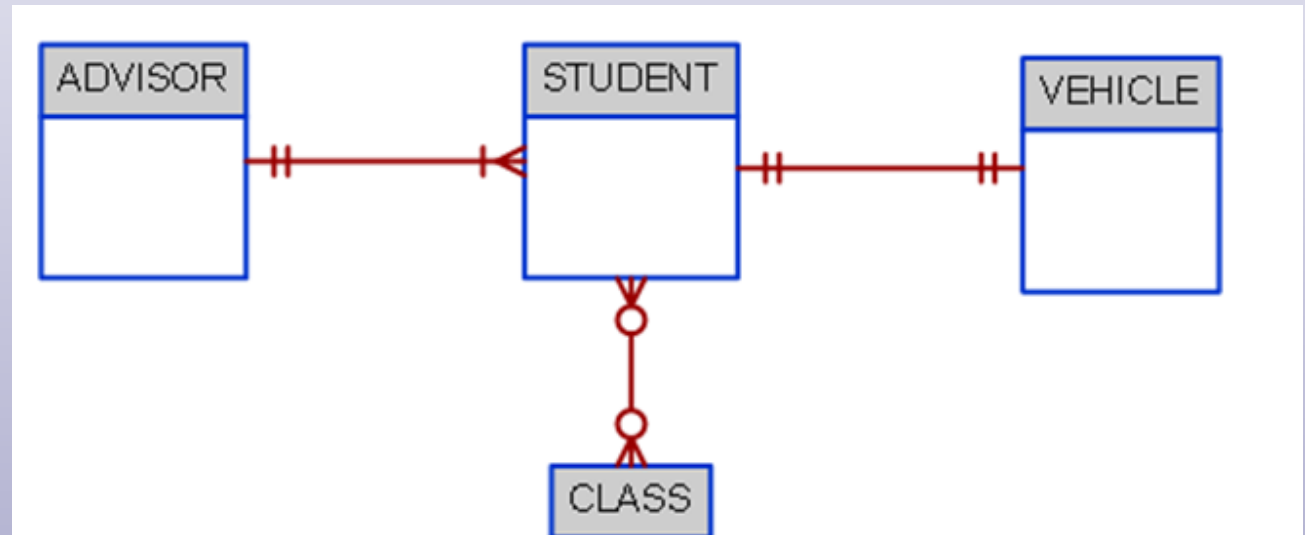


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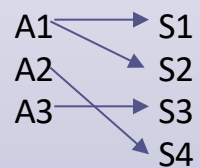
Consider the following example of data related together among the ADVISOR and STUDENT entities.
Show all the data of the remaining relationship diagram.

ADVISOR – STUDENT

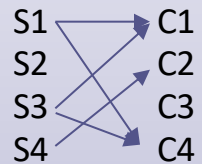
A1 → S1
A2 → S2
A3 → S3
A3 → S4



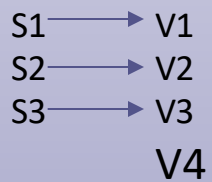
ADVISOR – STUDENT



STUDENT – CLASS



STUDENT – VEHICLE



COURSE – CLASS

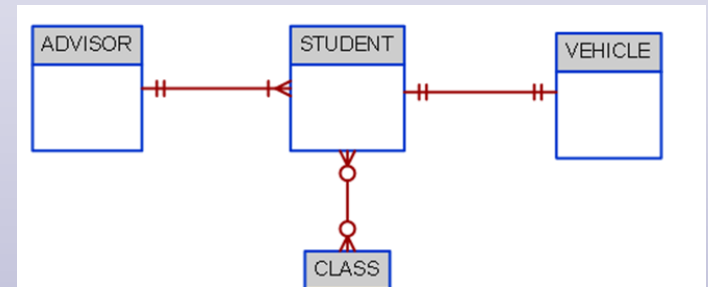
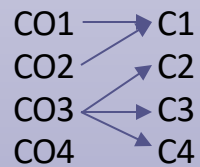
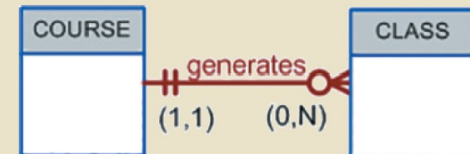


FIGURE 4.13 CLASS IS OPTIONAL TO COURSE



Relationship Degree

- Indicates the number of entities or participants associated with a relationship
- **Unary relationship:** Association is maintained within a single entity
 - **Recursive relationship:** Relationship exists between occurrences of the same entity set
- **Binary relationship:** Two entities are associated
- **Ternary relationship:** Three entities are associated

Figure 4.15 - Three Types of Relationship Degree

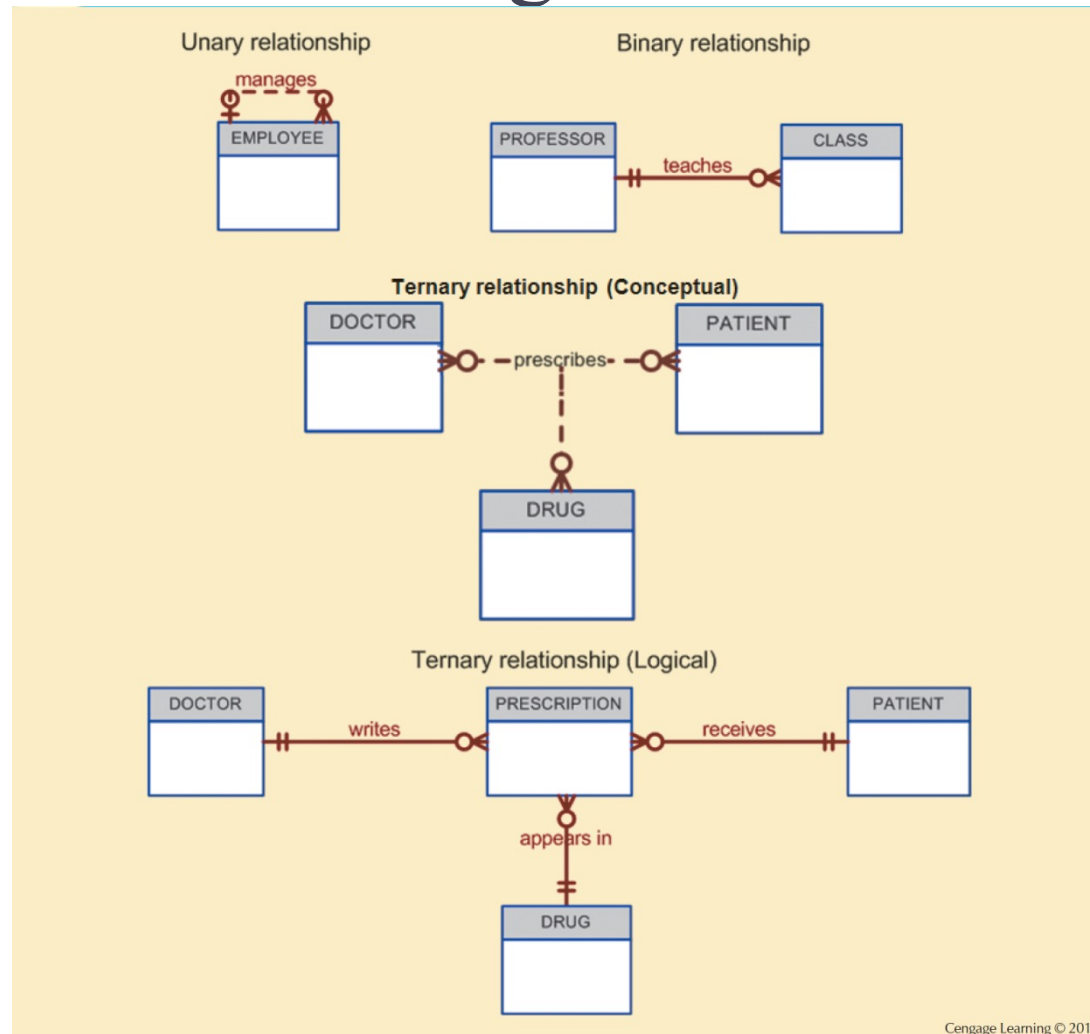
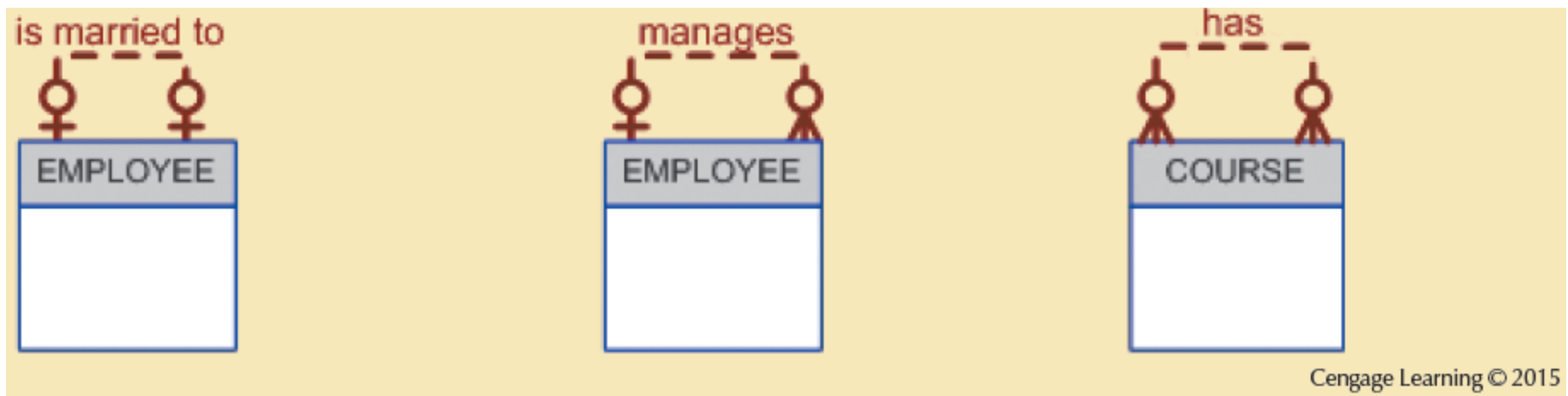


Figure 4.17 - An ER Representation of Recursive Relationships



Associative Entities

- Also known as composite or bridge entities
- Used to represent an M:N relationship between two or more entities
- Is in a 1:M relationship with the parent entities
 - Composed of the primary key attributes of each parent entity
- May also contain additional attributes that play no role in connective process

Figure 4.23 - Converting the M:N Relationship into Two 1:M Relationships

Table name: STUDENT

Database name: Ch04_CollegeTry

STU_NUM	STU_LNAME
321452	Bowser
324257	Smithson

Table name: ENROLL

CLASS_CODE	STU_NUM	ENROLL_GRADE
10014	321452	C
10014	324257	B
10018	321452	A
10018	324257	B
10021	321452	C
10021	324257	C

Table name: CLASS

CLASS_CODE	CRS_CODE	CLASS_SECTION	CLASS_TIME	ROOM_CODE	PROF_NUM
10014	ACCT-211	3	TTh 2:30-3:45 p.m.	BUS252	342
10018	CIS-220	2	MWTF 9:00-9:50 a.m.	KLR211	114
10021	QM-261	1	MWTF 8:00-8:50 a.m.	KLR200	114

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Figure 4.25 - A Composite Entity in an ERD

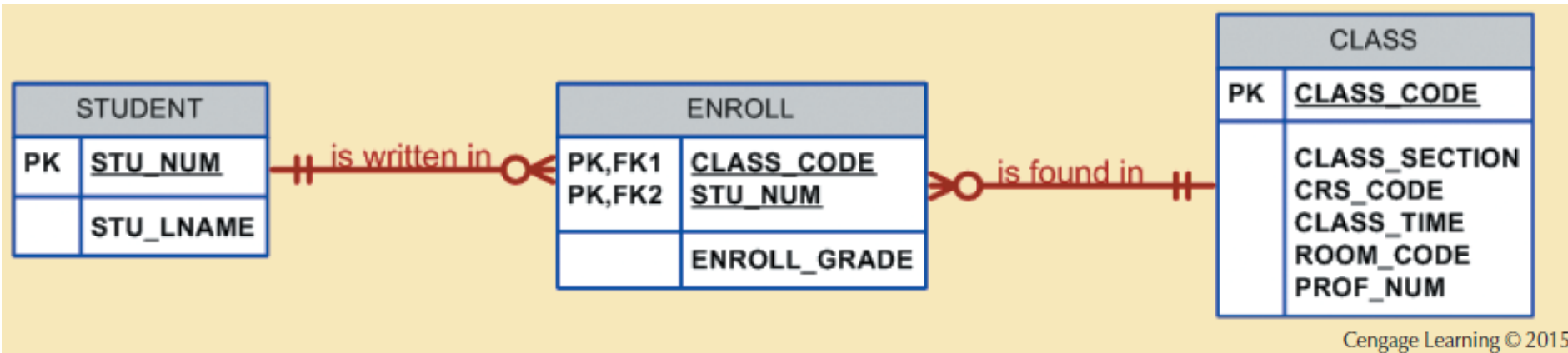
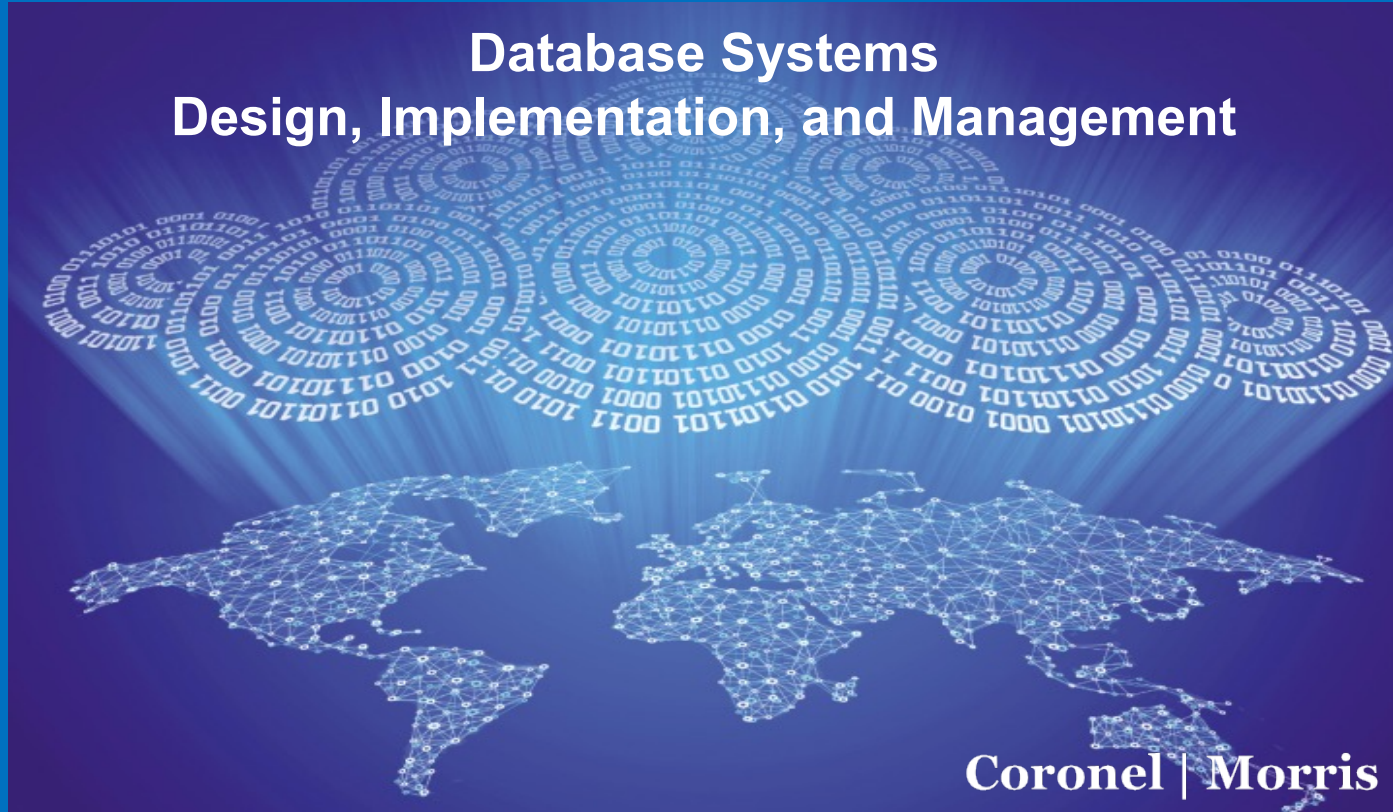


Table 4.4 - Components of the ERM

ENTITY	RELATIONSHIP	CONNECTIVITY	ENTITY
SCHOOL	operates	1:M	DEPARTMENT
DEPARTMENT	has	1:M	STUDENT
DEPARTMENT	employs	1:M	PROFESSOR
DEPARTMENT	offers	1:M	COURSE
COURSE	generates	1:M	CLASS
PROFESSOR	is dean of	1:1	SCHOOL
PROFESSOR	chairs	1:1	DEPARTMENT
PROFESSOR	teaches	1:M	CLASS
PROFESSOR	advises	1:M	STUDENT
STUDENT	enrolls in	M:N	CLASS
BUILDING	contains	1:M	ROOM
ROOM	is used for	1:M	CLASS
<i>Note: ENROLL is the composite entity that implements the M:N relationship "STUDENT enrolls in CLASS."</i>			

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Database Systems Design, Implementation, and Management



Chapter 3

The Relational Database Model

Learning Objectives

- In this chapter, one will learn:
 - That the relational database model offers a logical view of data
 - About the relational model's basic component: relations
 - That relations are logical constructs composed of rows (tuples) and columns (attributes)
 - That relations are implemented as tables in a relational DBMS

Learning Objectives

- In this chapter, one will learn:
 - About relational database operators, the data dictionary, and the system catalog
 - How data redundancy is handled in the relational database model
 - Why indexing is important

A Logical View of Data

- Relational database model enables logical representation of the data and its relationships
- Logical simplicity yields simple and effective database design methodologies
- Facilitated by the creation of data relationships based on a logical construct called a relation

Table 3.1 - Characteristics of a Relational Table

1	A table is perceived as a two-dimensional structure composed of rows and columns.
2	Each table row (tuple) represents a single entity occurrence within the entity set.
3	Each table column represents an attribute, and each column has a distinct name.
4	Each intersection of a row and column represents a single data value.
5	All values in a column must conform to the same data format.
6	Each column has a specific range of values known as the attribute domain .
7	The order of the rows and columns is immaterial to the DBMS.
8	Each table must have an attribute or combination of attributes that uniquely identifies each row.

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Column = attribute

ID	Name	HireDate	Salary	Department_id
1234	A	21-FEB-1995	5000	10
1235	B	12-MAR-1990	10000	20
		Row = สิ่งที่อยู่ในตาราง และประกอบด้วยข้อมูลจากหลาย ๆ column		

Table = ข้อมูลทั้งหมดที่รวมอยู่ใน column และใน row

ID	Name	HireDate	Salary	Department_id
1234	A	21-FEB-1995	5000	10
1235	B	12-MAR-1990	10000	20
1236	C	5000	21-FEB-1995	ABC

Keys

- Consist of one or more attributes that determine other attributes
- Used to:
 - Ensure that each row in a table is uniquely identifiable
 - Establish relationships among tables and to ensure the integrity of the data
- **Primary key (PK):** Attribute or combination of attributes that uniquely identifies any given row

ID	Name	HireDate	Salary	Department_id
1234	A	21-FEB-1995	5000	10
1235	B	12-MAR-1990	10000	20
1234	A	21-FEB-1995	5000	10
1234	A	21-FEB-1995	5000	10

Video: <https://www.studytonight.com/dbms/database-key.php>

Determination

- State in which knowing the value of one attribute makes it possible to determine the value of another
- Is the basis for establishing the role of a key
- Based on the relationships among the attributes

$A \rightarrow B$ อ่านว่า A determine B

B เป็น functional dependencies กับ A



StudentID	CourseID	EnrollDate	Credit	ProfessorID
S1234	Math	21-FEB-1995	3	10
S1234	Science	12-MAR-1995	3	20
S1234	Art	21-FEB-1995	3	10
S1235	Art	12-MAR-1995	3	10
S1236	Science	12-MAR-1995	3	20

StudentID → CourseID, EnrollDate, Credit, ProfessorID

CourseID → StudentID, EnrollDate, Credit, ProfessorID

StudentID, CourseID → EnrollDate, Credit, ProfessorID

Dependencies

- **Functional dependence:** Value of one or more attributes determines the value of one or more other attributes
 - **Determinant:** Attribute whose value determines another
 - **Dependent:** Attribute whose value is determined by the other attribute
- **Full functional dependence:** Entire collection of attributes in the determinant is necessary for the relationship



A	B	C
1	2	3
2	2	1
2	1	3

Super key คือ attribute ใดก็ตามที่กำหนดให้เป็น key แล้วสามารถบอกค่า attribute อื่น ๆ ได้เพียงค่าเดียว
จำนวน super key ที่เป็นไปได้ คือ 2^n เมื่อ n คือ จำนวน column และการกำหนด super key นั้นจะกำหนดจากหลักการ
determination

ดังนั้นจากตัวอย่างด้านบนนี้ จะมี list ของ super key เป็นไปได้ คือ

$A \rightarrow B, C$ คือ การทราบค่า A แล้วสามารถบอกค่า B และ C ได้เพียงค่าเดียว

$B \rightarrow A, C$

$C \rightarrow A, B$

$A, B \rightarrow C$ คือ การทราบค่า A และ B แล้วสามารถบอกค่า C ได้เพียงค่าเดียว

$A, C \rightarrow B$

$B, C \rightarrow A$

$A, B, C \rightarrow$

$\{ \} = \text{ค่า null}$



A	B	C
1	2	3
2	2	1
3	3	3

Super key คือ attribute ใดก็ตามที่กำหนดให้เป็น key แล้วสามารถบอกค่า attribute อื่น ๆ ได้เพียงค่าเดียว
จำนวน super key ที่เป็นไปได้ คือ 2^n เมื่อ n คือ จำนวน column และการกำหนด super key นั้นจะกำหนดจากหลักการ
determination

ดังนั้นจากตัวอย่างด้านบนนี้ จะมี list ของ super key เป็นไปได้ คือ

A $\rightarrow$ B, C
B $\rightarrow$ A, C
C $\rightarrow$ A, B
A, B $\rightarrow$ C
A, C $\rightarrow$ B
B, C $\rightarrow$ A
A, B, C $\rightarrow$
{ } = ค่า null

Types of Keys

- **Composite key:** Key that is composed of more than one attribute
- **Key attribute:** Attribute that is a part of a key
- **Entity integrity:** Condition in which each row in the table has its own unique identity
 - All of the values in the primary key must be unique
 - No key attribute in the primary key can contain a null

Types of Keys

- **Null:** Absence of any data value that could represent:
 - An unknown attribute value
 - A known, but missing, attribute value
 - A inapplicable condition
- **Referential integrity:** Every reference to an entity instance by another entity instance is valid

Table 3.3 - Relational Database Keys

KEY TYPE	DEFINITION
Superkey	An attribute or combination of attributes that uniquely identifies each row in a table
Candidate key	A minimal (irreducible) superkey; a superkey that does not contain a subset of attributes that is itself a superkey
Primary key	A candidate key selected to uniquely identify all other attribute values in any given row; cannot contain null entries
Foreign key	An attribute or combination of attributes in one table whose values must either match the primary key in another table or be null
Secondary key	An attribute or combination of attributes used strictly for data retrieval purposes

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Specify the super key, candidate key, primary key

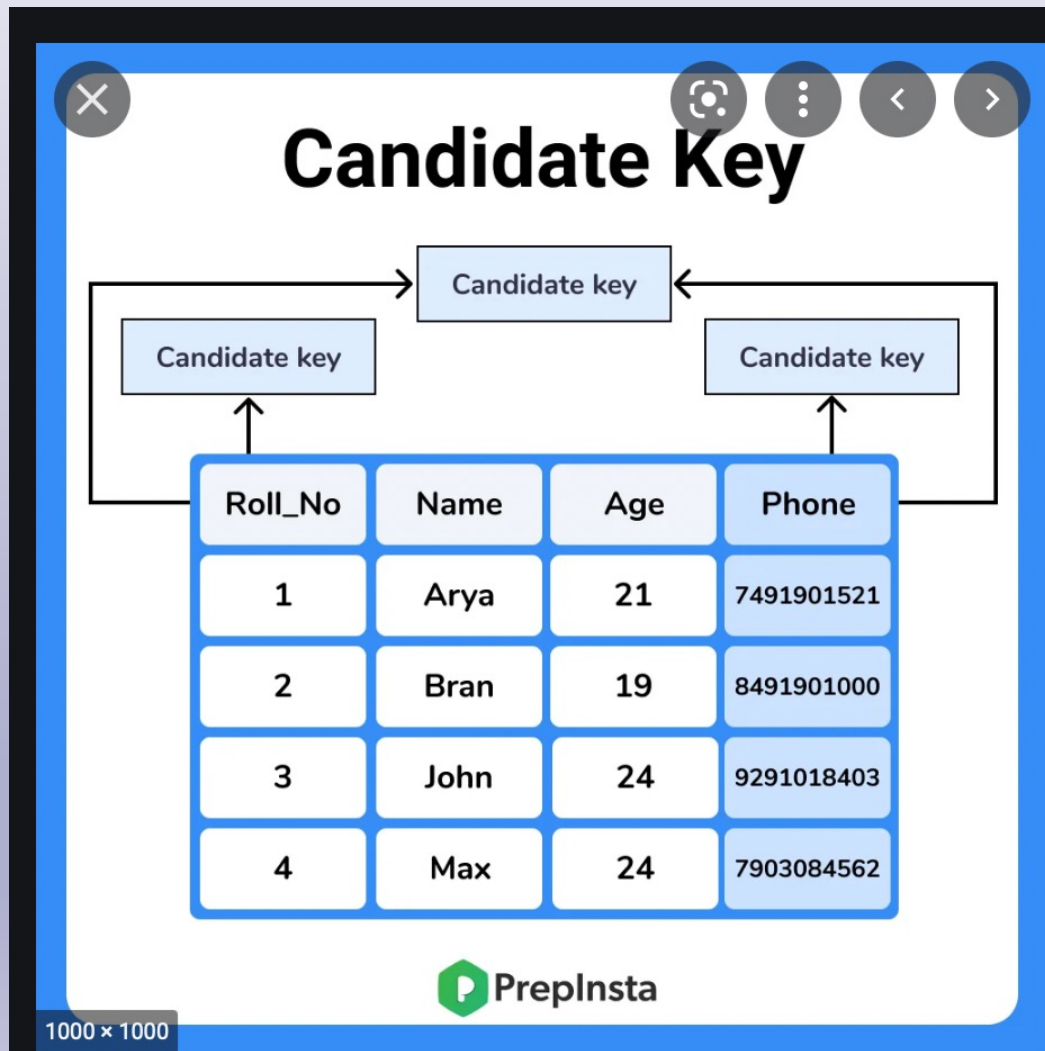


Figure 3.2 - An Example of a Simple Relational Database

Table name: PRODUCT

Primary key: PROD_CODE

Foreign key: VEND_CODE

Database name: Ch03_SaleCo

PROD_CODE	PROD_DESCRIPT	PROD_PRICE	PROD_ON_HAND	VEND_CODE
001278-AB	Claw hammer	12.95	23	232
123-21UUY	Houselite chain saw, 16-in. bar	189.99	4	235
QER-34256	Sledge hammer, 16-lb. head	18.63	6	231
SRE-657UG	Rat-tail file	2.99	15	232
ZZX/3245Q	Steel tape, 12-ft. length	6.79	8	235

link

Table name: VENDOR

Primary key: VEND_CODE

Foreign key: none

VEND_CODE	VEND_CONTACT	VEND_AREACODE	VEND_PHONE
230	Shelly K. Smithson	608	555-1234
231	James Johnson	615	123-4536
232	Annelise Crystall	608	224-2134
233	Candice Wallace	904	342-6567
234	Arthur Jones	615	123-3324
235	Henry Ortozo	615	899-3425

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Integrity Rules

Entity Integrity	Description
Requirement	All primary key entries are unique, and no part of a primary key may be null
Purpose	Each row will have a unique identity, and foreign key values can properly reference primary key values
Example	No invoice can have a duplicate number, nor it can be null

Integrity Rules

Entity Integrity	Description
Requirement	A foreign key may have either a null entry or a entry that matches a primary key value in a table to which it is related
Purpose	It is possible for an attribute not to have a corresponding value but it is impossible to have an invalid entry It is impossible to delete row in a table whose primary keys has mandatory matching foreign key values in another table
Example	It is impossible to have invalid sales representative number

Figure 3.3 - An Illustration of Integrity Rules

Table name: CUSTOMER

Database name: Ch03_InsureCo

Primary key: CUS_CODE

Foreign key: AGENT_CODE

CUS_CODE	CUS_LNAME	CUS_FNAME	CUS_INITIAL	CUS_RENEW_DATE	AGENT_CODE
10010	Ramas	Alfred	A	05-Apr-2014	502
10011	Dunne	Leona	K	16-Jun-2014	501
10012	Smith	Kathy	W	29-Jan-2015	502
10013	Olowski	Paul	F	14-Oct-2014	502
10014	Orlando	Myron		28-Dec-2014	501
10015	O'Brian	Amy	B	22-Sep-2014	503
10016	Brown	James	G	25-Mar-2015	502
10017	Williams	George		17-Jul-2014	503
10018	Farriss	Anne	G	03-Dec-2014	501
10019	Smith	Olette	K	14-Mar-2015	503

Table name: AGENT (only five selected fields are shown)

Primary key: AGENT_CODE

Foreign key: none

AGENT_CODE	AGENT_AREACODE	AGENT_PHONE	AGENT_LNAME	AGENT_YTD_SLS
501	713	228-1249	Alby	132735.75
502	615	882-1244	Hahn	138967.35
503	615	123-5589	Okon	127093.45

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Relationships within the Relational Database

- 1:M relationship - Norm for relational databases
- 1:1 relationship - One entity can be related to only one other entity and vice versa
- Many-to-many (M:N) relationship - Implemented by creating a new entity in 1:M relationships with the original entities
 - **Composite entity (Bridge or associative entity):**
Helps avoid problems inherent to M:N relationships, includes the primary keys of tables to be linked

Figure 3.21 - The 1:1 Relationship between PROFESSOR and DEPARTMENT

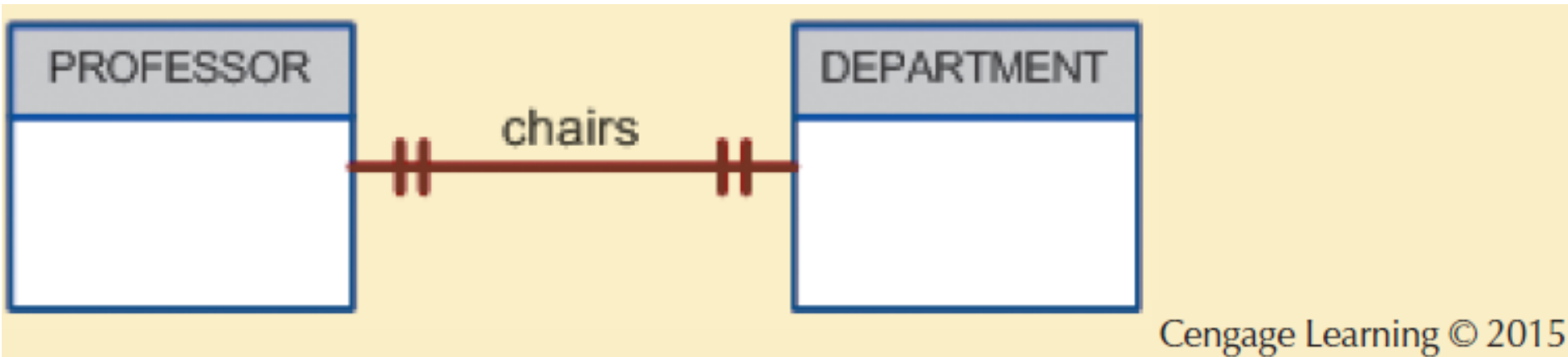


Figure 3.26 - Changing the M:N Relationship to Two 1:M Relationships

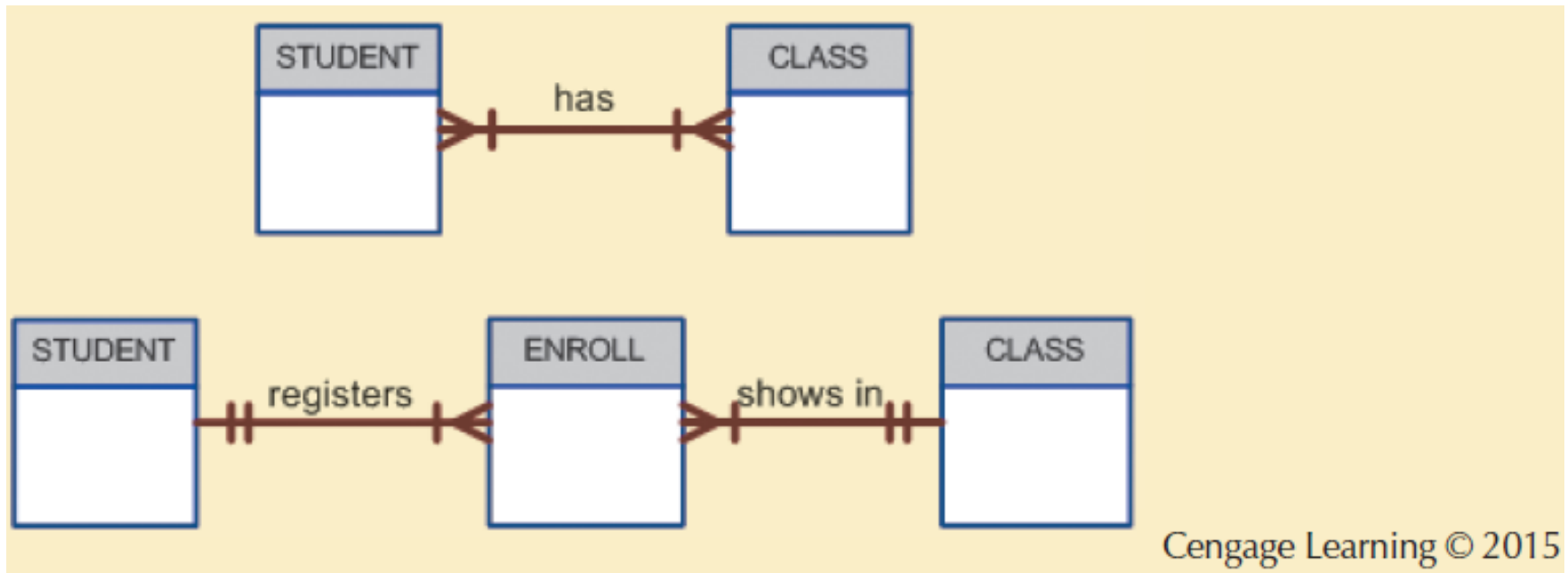
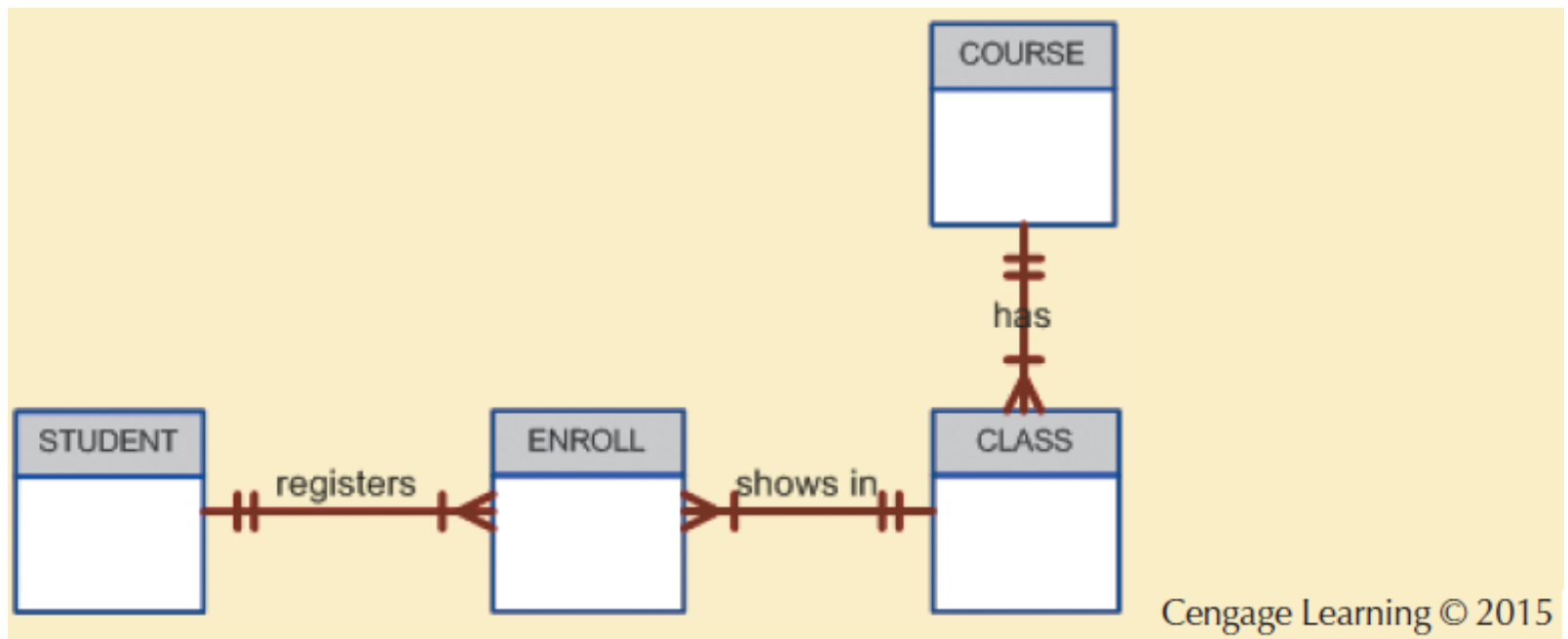


Figure 3.27 - The Expanded ER Model



Database Design Challenges: Conflicting Goals

Database design must conform to design standards

Need for high processing speed may limit the number and complexity of logically desirable relationships

Need for maximum information generation may lead to loss of clean design structures and high transaction speed

Figure 4.38 - Various Implementations of the 1:1 Recursive Relationship

Table name: EMPLOYEE_V1

EMP_NUM	EMP_LNAME	EMP_FNAME	EMP_SPOUSE
345	Ramirez	James	347
346	Jones	Anne	349
347	Ramirez	Louise	345
348	Delaney	Robert	
349	Shapiro	Anton	346

Database name: Ch04_PartCo

First implementation

Table name: EMPLOYEE

EMP_NUM	EMP_LNAME	EMP_FNAME
345	Ramirez	James
346	Jones	Anne
347	Ramirez	Louise
348	Delaney	Robert
349	Shapiro	Anton

Table name: MARRIED_V1

EMP_NUM	EMP_SPOUSE
345	347
346	349
347	345
349	346

Second implementation

Table name: MARRIAGE

MAR_NUM	MAR_DATE
1	04-Mar-03
2	02-Feb-99

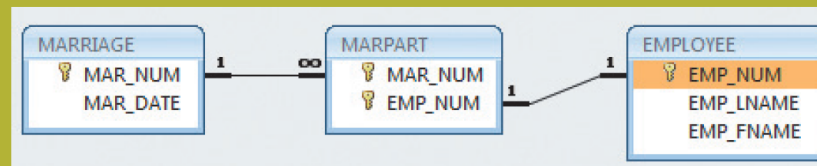
Table name: MARPART

MAR_NUM	EMP_NUM
1	345
1	347
2	346
2	349

Table name: EMPLOYEE

EMP_NUM	EMP_LNAME	EMP_FNAME
345	Ramirez	James
346	Jones	Anne
347	Ramirez	Louise
348	Delaney	Robert
349	Shapiro	Anton

The relational diagram for the third implementation



Third implementation